COGS 9: Introduction to Data Science
Assignment #3: Machine Learning (9pts)
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Introduction

In this assignment, you will explore an interactive article from Parametric Press titled "The Myth of the Impartial Machine." The piece explains how bias can enter machine learning systems through data, model design, and real-world feedback loops. You will use the article's interactive widgets to run short simulations and collect evidence, then respond to questions using what you observe.

What you'll be doing

1. Go to: https://parametric.press/issue-01/the-myth-of-the-impartial-machine/
2. Read the *entire* piece first (without answering yet) to understand the big picture.
    - If you are new to machine learning, the article may make more sense after our first ML class discussion.
3. Go back through the article and complete the questions below as you encounter the relevant interactive widgets.
4. When finished, download your completed work as a PDF and submit it on Gradescope.

Question 1 (1 pt)

In a few sentences, describe the relationship between machine learning and artificial intelligence.

Machine learning is a subfield of artificial intelligence. Artificial intelligence (AI) is the broader goal of creating systems that can perform tasks that normally require human intelligence, such as recognizing patterns or making decisions. Machine learning is one of the main methods used to achieve this goal, it allows computers to learn patterns from data and improve their performance without being explicitly programmed for every task.

In other words, AI is the overall concept of intelligent machines, while machine learning is a specific approach that uses data and statistical models to build those intelligent systems.

Question 2 (2 pt)

Background: We often can't collect data from an entire population, so we take a *sample* instead. This can introduce bias or error.

In the interactive widget **"Sampling Errors Can Lead to Biased Models,"** do the following:

- Collect **10 samples**, each with **sample size = 10**.
- Record the **sample mean** for each sample in the table.

Sample	1	2	3	4	5	6	7	8	9	10
Sample Size	10	10	10	10	10	10	10	10	10	10
Sample mean	10000	8500	10400	9000	8000	10700	13400	9300	12900	8900

Then answer:

A. Interpret the results of your 10 samples.
   a. What do you notice about how the sample means compare to each other and to the **true population mean**?
   b. Explain how this relates to sampling bias and/or sampling error.

B. In a few sentences, compare **sampling errors** vs. **non-sampling errors**.
   a. How are they different?
   b. Give an example of each.

A.

a. Sample means are all different, ranging from 8000 to 13400. Even though every sample has the same size 10, the averages still change because each sample includes different data points. Most of the values are still around 10,000, so they are somewhat close to the true population mean. This shows that sample averages can vary, but they usually stay in the same general range.

b. It's an example of **sampling error**. Since we are only looking at small samples instead of the whole population, the results will naturally change from sample to sample. The differences between the sample means and the true population mean happen because of random variation in the data.

B.

a. Sampling errors happen because we are using a sample instead of the whole population. The results depend on which data points end up in the sample. Non-sampling errors happen because of mistakes or bias when collecting or measuring the data.
b. An example of a sampling error is when a small sample gives a mean that is higher or lower than the real population mean. An example of a non-sampling error is if a survey only collects responses from one group of people and ignores others, which makes the results biased.

Question 3 (1 pt)

A. Briefly explain the sampling bias issue in the ImageNet dataset described in the article.
B. If you were in charge of improving the dataset, describe at least one concrete step you would take to reduce this bias.

The article explains that the ImageNet dataset has sampling bias because many images come from Western internet sources. This means some groups and cultures are overrepresented while others are missing, which can make models less accurate for certain groups.

To reduce this bias, I would collect more images from underrepresented regions and communities so the dataset better reflects the real world.

Question 4 (2 pt)

In the section "Algorithms can amplify bias found in data," try five different combinations of the bias and accuracy sliders.

For each trial, record:

- the bias setting (e.g., 70/30),
- the model accuracy (%),
- the result/outcome shown by the widget.

	Bias	Model Accuracy	Result
Example	*70/30*	*70%*	*Not incentivized to amplify bias*
1	90/10	80%	Model would be incentivized to amplify bias.
2	80/20	90%	Model would not be incentivized to amplify bias.
3	90/10	50%	Model would be incentivized to amplify bias

4	70/30	70%	Model **would not** be incentivized to amplify bias
5	100/0	80%	Model **would** be incentivized to amplify bias

A. Using your data, explain the relationship between **model bias** and **model accuracy**.
B. Under what conditions does the model appear **incentivized to amplify bias**, and why?

A. Results show that model accuracy alone does not determine whether the model amplifies bias. The **bias in the data distribution** also matters. When the dataset is very unbalanced (like **90/10 or 100/0**), the model can still have high accuracy while mostly predicting the majority group. This means a model can appear accurate but still produce biased outcomes.

B. Model is incentivized to amplify bias when the **data is highly imbalanced**, such as **90/10 or 100/0**. In these cases, the model can get higher accuracy by mostly predicting the majority group and ignoring the minority group. Because the model is optimized for accuracy, it learns patterns that favor the larger group, which increases bias.

Question 5 (1 pt)

In the **Feedback Loops Runaway** simulation:

- Set the number of crimes in **Precinct A = 4** and **Precinct B = 4** (equal crime levels).
- Run the simulation and watch how the system changes over time.

Explain what happens in the simulation. In your explanation, describe what the outcome suggests about the relationship between **reported crime**, **policing decisions**, and **feedback loops**.

When both Precinct A and B start with **4 crimes**, they begin with the same crime level. However, over time one precinct may receive **more police attention**, which leads to **more crimes being detected and reported** there.

This creates a **feedback loop**: more policing leads to more reported crime, which then causes even more policing in that area, even though the original crime levels were the same.

Question 6 (1 pt)

The article describes three methods people use to tackle bias. Choose **one** approach that you think is the most important.

In a few sentences:

A. Identify and describe the approach you chose (as presented in the article).
B. Critically evaluate it by explaining **pros and cons** (benefits and limitations).
    a. There is no single correct answer—this question is graded on thoughtful reasoning.

One approach discussed in the article is **improving the data used to train models**. This means collecting more representative data so that different groups, environments, and situations are included in the dataset.

A benefit of this approach is that better and more balanced data can help reduce bias in the model's predictions. If the dataset reflects the real world more accurately, the model is less likely to favor one group over another. However, a limitation is that it can be **difficult and expensive to collect truly representative data**, and some biases may still remain even after improving the dataset.

Question 7 (1 pt)

How do you think the issues discussed in this piece—**biased data, biased algorithms, and efforts to reduce bias**—will change over time? Do you think they will improve, stay the same, or get worse?

State your position and defend it with reasoning. There is no right or wrong answer; this is about critical thinking.

For each item, choose **one** prediction (**improve / stay the same / worsen**) and justify it using **at least one concrete reason** (example: incentives, regulation, technical advances, data collection practices, accountability, public pressure).

FILL IN BELOW WITH YOUR ANSWERS.

A. Biased data
    a. Prediction: Improve
    b. Justification (1-2 sentences): As awareness of bias increases, organizations will likely put more effort into collecting **more diverse and representative datasets**. Public pressure and research in AI fairness are pushing companies and researchers to pay more attention to how data is collected.

B. Biased algorithms
    a. Prediction:Stay the same
    b. Justification (1-2 sentences): Even if models improve technically, algorithms still depend on the data and objectives they are given. Since many systems are still optimized mainly for accuracy and efficiency, bias may continue to appear in different forms.
C. Efforts to reduce bias
    a. Prediction: Improve
    b. Justification (1-2 sentences): There is growing regulation, research, and public attention around AI fairness. Governments, universities, and tech companies are increasingly studying bias and developing tools to detect and reduce it, which will likely lead to better solutions over time.

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